

# A Comprehensive AI-Based Mental Health Monitoring Toward Nervous System Exhaustion

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**Abstract**— Mental health problems such as stress, anxiety, depression, and mental fatigue have become increasingly prevalent, creating a growing need for a reliable and timely monitoring system. Recent advancements in artificial intelligence have led to the development of various approaches that utilize physiological signals, wearable devices, social media data, and multimodal information for mental health assessment. This literature review examines recent studies employing traditional machine learning, deep learning, wearable sensing technologies, multimodal fusion, and transformer-based models for detecting mental health conditions. The reviewed studies are analyzed based on their methodologies, datasets, performance, strengths, and existing limitations. The analysis shows that while recent models have achieved considerable improvements in detection accuracy, several challenges remain, including limited dataset diversity, poor generalization across real-world environments, computational complexity, privacy concerns, and insufficient long-term monitoring of an individual's psychological state. These observations indicate the need for more comprehensive and continuous assessment frameworks that consider both physiological and behavioral changes over time. This review summarizes current research trends, identifies existing gaps, and discusses future directions that can contribute to the development of intelligent systems for Nervous System Exhaustion (NSE)\* assessment and early burnout risk identification.

**Keywords**—Mental Health Monitoring, Artificial Intelligence, Stress Detection, Anxiety Detection, Depression Detection, Mental Fatigue, Wearable Sensors, Deep Learning, Transformer Models, Nervous System Exhaustion (NSE), Literature Review.

## I. INTRODUCTION (HEADING 1)

The increasing prevalence of stress, anxiety, depression, and mental fatigue has made mental health monitoring an important area of research. These conditions significantly affect emotional well-being, cognitive performance, physical health, and overall quality of life. Recent advancements in wearable sensing technologies and artificial intelligence have enabled continuous monitoring of physiological signals, including electrocardiography (ECG), heart rate variability (HRV), photoplethysmography (PPG), and electrodermal

activity (EDA), along with behavioral data obtained from digital platforms. Conventional machine learning techniques such as Support Vector Machines (SVM), Random Forest (RF), and Decision Trees (DT), as well as deep learning models including Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, multimodal fusion models, and transformer-based architectures, have demonstrated promising performance in automated mental health assessment.

Despite these advancements, several challenges continue to limit the practical deployment of existing systems. Many studies rely on small datasets collected in controlled environments, resulting in limited generalization to real-world scenarios. Furthermore, issues such as noisy physiological signals, high computational complexity, limited model interpretability, privacy concerns, and the emphasis on detecting only the current mental state remain unresolved. This literature review presents a comprehensive analysis of recent AI-based mental health monitoring approaches by examining their methodologies, datasets, strengths, and limitations. It also identifies existing research gaps and highlights the need for intelligent systems capable of supporting long-term mental health assessment and the early detection of Nervous System Exhaustion (NSE).

## II. LITERATURE SURVEY

### Review of Existing Papers:

#### 1. Calibration of Transformer-Based Models for Identifying Stress and Depression in Social Media(2024)

This paper proposed an enhanced transformer-base framework for detecting stress and depression from social media posts by integrating linguistic features with BERT and Mental BERT models. The approach combined LIWC NRC Emotion Lexicon, LDA topic modelling, and Top2Vec features using a multimodal adaption gate, while label smoothing was applied to improve model calibration and prediction reliability. Experimental results demonstrated improved classification accuracy and better calibration compared to conventional transformer-based architecture

required significant computational resources and lacked explainability for practical mental health application.

## *2. Analysis of Computer Vision-based Physiological Indicators for operator Fatigue Detection(2025)*

This paper proposed a computer vision-based fatigue detection framework using a Tabular Transformer model to classify operator fatigue from physiological indicators extracted through facial analysis. To improve model transparency, the authors incorporated SHAP and permutation Importance techniques, allowing important features to be identified and interpreted. The proposed framework achieved effective fatigue classification while enhancing explainability, making it suitable for safety-critical applications. However, the model requires high computational resources when multiple deep learning and explainability techniques are executed simultaneously, limiting real-time deployment on resource-constrained systems.

## *3. Assessing Occupational Burnout in Psychiatric Professionals Using Multimodal Emotion Recognition Methods (2026)*

This Study developed a multimodal emotion recognition framework to assess occupational burnout among psychiatric professionals by combining physiological signals such as ECG and PPG with Facial video features. The integration of multiple data sources improved burnout detection accuracy and provided a more comprehensive understanding of emotional and physiological conditions. Experimental results showed that multimodal fusion significantly enhanced prediction performance compared to single-modal approaches. However, the model's performance declined considerably when individual physiological or facial modalities were used independently, demonstrating its dependence on multimodal data fusion.

## *4. Deep Learning Approaches for Stress Detection: A Survey (2025)*

This survey reviewed recent deep learning and multimodal approaches for automated stress detection using physiological, behavioural, and contextual data. The study discussed techniques including CNNs, LSTMs, Transformers, and multimodal fusion models, highlighting their ability to achieve high stress detection accuracy through automatic feature learning. It also examined commonly used physiological signals and wearable sensing technologies for continuous monitoring. However, the survey identified several challenges, including limited benchmark datasets, inconsistent labeling methods, lack of standardized evaluation protocols, and poor modal generalization across different environments and applications.

## *5. Psychological Stress Prediction and Prevention in Higher Education Teachers Using a Bi-LSTM Based Multilayer Framework(2025)*

This paper introduced a Bi-LSTM-based multilayer framework for predicting psychological stress among higher education teachers using sequential multimodal time-series data. The proposed model effectively captured temporal relationships in physiological and behavioural signals, resulting in strong stress prediction performance. The framework demonstrated the potential of deep sequential learning for early stress detection and mental health monitoring in educational environments. However, the experimental validation was conducted using limited datasets collected under specific conditions, restricting the model's ability to generalize across different institutions and real-world settings.

## *6. Physiological Sensing and Machine learning Approaches Towards Anxiety Detection: A Systematic Review(2025)*

This Systematic review analyzed recent development in physiological sensing and machine learning techniques for anxiety detection using signals such as EDA, HRV, ECG, PPG, respiration, and EEG. The review found that physiological sensing combined with machine learning provides reliable and objective anxiety detection, with multimodal approaches generally outperforming single-sensors systems. It also highlighted the increasing role of wearable devices in continuous mental health monitoring. However, heterogeneous datasets, inconsistent labeling methods, and limited workplace validation remain significant challenges for practical implementation.

## *7. Causal Inference of Construction Environmental Factors on Worker Health: Stress Probability Modeling via Gaussian Process Networks (2025)*

This paper proposed a stress probability prediction model using Gaussian Process Networks (GPN) and Gaussian Process Regression (GPR) to analyze the influence of construction site environmental factors on worker stress. The proposed framework effectively modelled uncertainty and captured complex relationships between environmental conditions and stress probability, supporting proactive occupational health monitoring. The results demonstrated the usefulness of probabilistic modelling for stress prediction in construction environments. However, the study was conducted using a relatively small participant group, limiting the generalizability of the findings to larger populations.

## *8. Extending Stress Detection Reproducibility to Consumer (2025)*

This paper investigated the reproducibility of stress detection models across different consumer wearable devices using physiological signals such as Heart Rate Variability (HRV) and Electrodermal Activity (EDA). The authors compared research-grade and commercial wearable sensors to evaluate stress detection performance under various conditions. The results showed that combining HRV and EDA significantly improved stress prediction and

demonstrated the potential of consumer wearables for continuous real-world monitoring. However, differences in device hardware and sensor compatibility affected model generalization and reproducibility across different wearable platforms

#### 9. *Toward Camara-PRV Based Early Warning in Hospital ICU: A Pilot Study (2025)*

This paper presented a camera-based remote physiological monitoring system for early stress and health assessment in intensive care units (ICUs). The proposed framework utilized remote Photoplethysmography (rPPG) to extract physiological signals from facial videos and employed machine learning algorithms including Support Vector Machine (SVM), Random Forest (RF), and K-Nearest Neighbor (KNN) for classification. The system enabled non-contact monitoring without requiring wearable sensors, making it suitable for continuous patient observation. However, the achieved classification accuracy was moderate, indicating that further optimization and clinical validation are required before practical deployment in hospital environments.

#### 10. *Ambient Stress Detection for Open Environment Using Hybrid Model (2024)*

This paper proposed a hybrid stress detection model by combining Support Vector Machine (SVM) and Random Forest algorithms for monitoring stress in open and dynamic environments. The framework processed physiological sensor data to classify stress levels while improving prediction accuracy through hybrid machine learning techniques. Experimental evaluation demonstrated reliable real-time stress monitoring under different environmental conditions. The proposed approach offers potential for workplace and outdoor health monitoring applications. However, the model remained sensitive to sensor noise, environmental disturbances, and individual physiological differences, which affected its overall robustness.

#### 11. *Perceived Stress Prediction among Working Employees Based on CAT-DCNN Technique (2024)*

This paper proposed a CAT-DCDNN framework for predicting perceived stress among working employees using physiological data collected from wearable devices. The system incorporated preprocessing, feature selection, and a Deep Convolutional Neural Network (DCNN) optimized with the Cat Swarm Optimization (CAT) algorithm to improve classification performance. Experimental results showed that the optimized model achieved higher stress prediction accuracy than conventional machine learning approaches while effectively extracting important stress-related features. However, the wearable devices used for data collection were affected by sensor limitations, and battery constraints, which may reduce the system's effectiveness during long-term real-world monitoring.

#### 12. *Personalized Stress Monitoring Using Wearable Sensors in Everyday Settings (2023)*

This study presented a personalized stress monitoring system using wearable sensors to collect physiological signals such as heart rate (HR), Heart rate variability (HRV), during everyday activities. The proposed framework

employed Active Learning along with machine learning classifiers including random forest and SVM to build user-specific stress prediction models while minimizing manual data labeling. Experimental results demonstrated improved personalized stress detection compared to generic models and highlighted the advantages of adaptive monitoring. However, motion artifacts, noisy physiological signals, and variations in daily activities reduced the model's accuracy under real-world conditions.

#### 13. *Stress Detection Using Context-Aware Sensor Fusion from Wearable Devices (2023)*

This paper introduced the SELF-CARE framework, a context-aware sensor fusion approach for stress detection using wearable devices. The proposed model combines multiple physiological signals through an ensemble machine learning architecture while adapting to different sensing conditions by identifying sensor noise contexts. The framework achieved high stress classification accuracy on the WESAD dataset and demonstrated the effectiveness of selective sensor fusion for wearable healthcare applications. However, the model was evaluated on a relatively small benchmark dataset and requires validation across a larger and more diverse population for practical deployment.

#### 14. *Review on Stress Detection and Management Techniques Using Non-EEG Sensors (2022)*

This review paper presented a comprehensive overview of stress detection and management techniques using physiological signals obtained from ECG, EMG, GSR, HRV, Skin conductance, and other non-EEG sensors. It compared several machine learning algorithms including Random Forest, Support Vector Machine (SVM), K-Nearest Neighbor (KNN), Artificial Neural Networks (ANN), and Bayesian Networks for stress classification. The review also discussed IoT-enabled healthcare systems and cloud-based stress monitoring applications. However, as a survey paper, it did not include experimental implementation, or performance validation of the reviewed techniques in real-world scenarios.

#### 15. *IoT-Driven GSR Stress Detection: Clinical, Physical, and Linguistic Innovations (2022)*

This paper proposed an IoT-enabled stress detection framework using Galvanic Skin Response (GSR) signals combined with deep learning techniques such as Convolutional Autoencoders, LSTM Autoencoders, and clustering algorithms. The system continuously monitored physiological responses to identify stress patterns and supported remote healthcare applications through IoT connectivity. Experimental results demonstrated effective stress recognition from physiological data and highlighted the benefits of automated feature extraction. However, the deep learning models lacked clinical interpretability and transparency, making it difficult to explain the reasoning behind stress predictions in healthcare settings.

#### 16. *Multi-Modal Acute Stress Recognition Using Off-the-Shelf Wearable Devices (2021)*

This paper developed a multimodal acute stress recognition framework using data collected from commercially available wearable devices. The proposed

system combined multiple physiological signals and employed Random Forest and Support Vector Machine (SVM) classifiers to improve stress recognition accuracy. By integrating information from different wearable sensors, the framework achieved better classification performance than single sensor approaches and demonstrated its suitability for real-time monitoring. However, synchronization issues between different wearable sensors occasionally affected data quality, leading to reduced recognition performance under certain conditions.

#### 17. *EmoCv: Towards Physiological Signals-Based Stress Detection (2020)*

This study proposed the EmoCv framework for stress detection using physiological signals and time-domain feature extraction techniques. The system utilized ensemble machine learning algorithms such as Gradient Boosting and Random Forest to classify stress levels from physiological measurements. Experimental evaluation demonstrated high classification accuracy and robust performance across different stress conditions, showing the effectiveness of ensemble learning for physiological signal analysis. However, the framework relied on relatively short observation windows, which increased the likelihood of false alarms and reduced stability during continuous real-time stress monitoring.

#### 18. *Stress Detection Using Deep Neural Networks (2020)*

This paper developed models for stress detection using physiological sensor data collected from wearable devices. The proposed approach employed a One-Dimensional Convolutional Neural Network(1D-CNN) and a Multilayer Perceptron (MLP) to automatically learn discriminative features without requiring manual feature engineering. Experimental results demonstrated high stress classification accuracy and showed that deep neural networks outperformed several traditional machine learning techniques. However, the models required large volumes of labelled training data and considerable computational resources, making deployment difficult in data-constrained and resource-limited environments.

#### 19. *Mental Fatigue Prediction Model Based on Multimodal Fusion (2019)*

This paper proposed a multimodal fusion framework for predicting mental fatigue by combining EEG signals, physiological measurements, and workload indicators. The system integrated information from multiple data sources to improve fatigue prediction accuracy and support continuous real-time monitoring of mental workload. Experimental results showed that multimodal fusion significantly enhanced prediction performance compared to single-modal approaches and provided more reliable fatigue assessment. However, individual physiological differences among users affected prediction consistency, indicating the need for personalized calibration to improve generalization across diverse populations.

#### 20. *Design and Implementation of a Real-time Stress Monitoring system with the help of ECG using MATLAB tool (2017)*

This paper presented a real-time ECG-based stress monitoring system developed using the MATLAB platform. The proposed framework analyzed ECG signals through peak detection and threshold matching algorithms to identify stress levels and generate immediate alerts. The system demonstrated a simple, low-cost, and efficient approach for continuous stress monitoring, making it suitable for basic healthcare applications. However, the use of fixed threshold values limited the model's adaptability to individual physiological differences, reducing its effectiveness for personalized stress assessment in real-world scenarios.

### III. COMPARATIVE ANALYSIS OF THE PAPERS

| SL No | PAPER                                                      | TECHNIQUES USED            | STRENGTHS                                 | LIMITATIONS                         |
|-------|------------------------------------------------------------|----------------------------|-------------------------------------------|-------------------------------------|
| 1     | Analysis of Computer Vision-Based Physiological Indicators | Tabular Transformer + SHAP | Good fatigue detection, explainable model | High computational cost             |
| 2     | Occupational Burnout Using Multimodal Emotion Recognition  | ECG/PPG + Facial Features  | Uses multimodal information               | Accuracy drops with single modality |
| 3     | Deep Learning Approaches for Stress Detection              | Multimodal Deep Learning   | High detection performance                | Small benchmark datasets            |
| 4     | Psychological Stress Prediction in Teachers                | Bi-LSTM Framework          | Handles sequential data well              | Limited real-world validation       |
| 5     | Anxiety Detection Systematic Review                        | ML/DL+ HRV + EDA           | Comprehensive analysis                    | Dataset heterogeneity               |
| 6     | Construction Worker Stress Modeling                        | Gaussian Process Network   | Models' uncertainty effectively           | Very small sample size              |
| 7     | Wearable Stress Detection Reproducibility                  | HRV + EDA + ML             | Suitable for wearable devices             | Device compatibility issues         |
| 8     | Camera-PRV Based Early Warning                             | rPPG + SVM/RF/KNN          | Non-contact monitoring                    | Moderate accuracy                   |
| 9     | Ambient Stress Detection                                   | SVM + Random Forest        | Real-time stress monitoring               | Sensitive to sensor noise           |
| 10    | Transformer Models for Stress Identification               | BERT + MentalBERT          | Improved classification performance       | Computationally expensive           |

|    |                                             |                              |                               |                                            |
|----|---------------------------------------------|------------------------------|-------------------------------|--------------------------------------------|
| 11 | Perceived Stress Prediction                 | CNN + Cat Swarm Optimization | High prediction accuracy      | Battery and sensor limitations             |
| 12 | Personalized Stress Monitoring              | Active Learning + RF/SVM     | Personalized predictions      | Performance varies in daily life           |
| 13 | Context-Aware Sensor Fusion                 | Sensor Fusion + RF           | Effective multimodal analysis | Small participant group                    |
| 14 | Review on Stress Detection Techniques       | RF, SVM, KNN, ANN            | Compares multiple methods     | Review-based, not practical implementation |
| 15 | IoT-Driven GSR Stress Detection             | CAE + LSTM Autoencoder       | Supports IoT monitoring       | Limited explainability                     |
| 16 | Multimodal Acute Stress Recognition         | RF + SVM Fusion              | Good multimodal performance   | Synchronization issues                     |
| 17 | EmoCv Physiological Stress Detection        | Ensemble Learning            | High accuracy                 | False alarms possible                      |
| 18 | Stress Detection using Deep Neural Networks | CNN + MLP                    | Learns features automatically | Requires large training data               |
| 19 | Mental Fatigue Prediction Model             | Multimodal Fusion            | Real-time prediction          | Individual variability affects results     |
| 20 | Real-Time Stress Monitoring System          | ECG + Threshold Algorithms   | Simple and real-time          | Less adaptive and flexible                 |

#### IV. RESEARCH GAP

The studies reviewed demonstrate considerable progress in the application of artificial intelligence for stress, anxiety, depression, and mental fatigue detection. Machine learning, deep learning, wearable sensing technologies, multimodal fusion, and transformer-based models have significantly improved the accuracy of mental health assessment. However, several limitations continue to restrict their effectiveness in real-world applications.

A common challenge observed across the literature is the reliance on small or controlled datasets, which limits the ability of models to generalize across diverse populations. Physiological signals obtained from wearable devices are often affected by noise and motion artifacts, while social media-based approaches may not accurately represent an individual's actual mental state. Furthermore, many existing methods focus primarily on identifying the current physiological condition instead of analyzing long-term physiological and behavioral changes that gradually develop over time.

In addition, issues such as limited model interpretability, computational complexity, privacy concerns, and insufficient

integration of multiple data sources remain largely unresolved. These challenges highlight the need for more robust and intelligent monitoring frameworks capable of continuous assessment using multimodal information. Such systems can support the early identification of Nervous System Exhaustion (NSE), enable timely intervention and reduce the risk of prolonged mental health complications and burnout.

#### V. FUTURE RESEARCH DIRECTIONS

The growing adoption of artificial intelligence has significantly advanced mental health monitoring; However, several challenges continue to provide opportunities for future research. One important direction is the development of models that can generalize across diverse populations and perform reliably in real-world environments rather than only under controlled experimental conditions. Expanding the availability of large, diverse, and publicly accessible datasets which also improve the robustness and reliability of future based mental health systems.

Future studies should focus on integrating multimodal data from physiological signals, wearable devices, behavioral patterns, and contextual information to obtain a more comprehensive understanding of an individual's mental health. Advance in sensor technology, cloud computing, and edge intelligence can further support continuous, real-time monitoring while reducing computational overhead. At the same time, improving model interpretability, ensuring data privacy, and developing energy-efficient solutions will be essential for increasing user trust and enabling large-scale deployment in healthcare settings.

Another promising research direction for immediate mental health conditioned to understand long-term physiological and behavioral changes. Rather than identifying only the current level of stress or depression, future intelligent systems should continuously monitor recovery patterns and cumulative changes over time. Such an approach could support the early assessment of Nervous System Exhaustion (NSE), facilitate timely intervention, and contribute to more personalized, preventative and sustainable mental healthcare solutions.

#### VI. CONCLUSION

This literature review presented a comprehensive analysis of recent advancements in artificial intelligence for mental health monitoring, with a focus on stress, anxiety, depression, and mental fatigue detection. The reviewed studies demonstrate that machine learning, deep learning, wearable sensing technologies, multimodal fusion, and transformer-based models have significantly improved the accuracy and efficiency of mental health assessment. Each approach offers distinct advantages while also presenting challenges related to data quality, model generalizations, interpretability, privacy, and long-term deployment.

The comparative analysis reveals that although existing techniques are effective in identifying an individual's current psychological condition, they often lack the ability to continuously monitor long-term physiological and behavioral changes. This highlights the need for more reliable, interpretable, and multimodal frameworks capable of supporting continuous mental health assessment in real-world environments.

Overall, this review provides a consolidated understanding of current research trends, identifies key limitations in existing studies, and outlines potential directions for future work. Advancing research toward continuous assessment of Nervous System Exhaustion (NSE) can contribute to earlier identification of burnout risk and support the development of more personalized and preventive mental healthcare solutions.

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